When applied with $a_v = 2^{-m_v}$, this yields

$$\frac{1}{n} \geq \frac{1}{n} \sum_{v=1}^{n} 2^{-m_v} \geq \Big( \prod_{v=1}^{n} 2^{-m_v} \Big)^{1/n} = 2^{-\frac{1}{n} \sum_{v=1}^{n} m_v},$$

from which $2^{\frac{1}{n} \sum_{v=1}^{n} m_v} \geq n$, and hence, also $\sum_{v=1}^{n} m_v \geq n \log_2 n$ follows. □

## 3.5 2-colorable families

Let $\mathcal{F}$ be a family of subsets of some finite set. Can we color the elements of the underlying set in red and blue so that no member of $\mathcal{F}$ will be monochromatic? Such families are called 2-*colorable*.

Recall that a family is *k-uniform* if each member has exactly $k$ elements.

**Theorem 3.4** (Erdős 1963b)**.** *Every k-uniform family with fewer than* $2^{k-1}$ *members is* 2*-colorable.*

*Proof.* Let $\mathcal{F}$ be an arbitrary $k$-uniform family of subsets of some finite set $X$. Consider a random 2-coloring obtained by coloring each point independently either red or blue, where each color is equally likely. Informally, we have an experiment in which a fair coin is flipped to determine the color of each point. For a member $A \in \mathcal{F}$, let $X_A$ be the indicator random variable for the event that $A$ is monochromatic. So, $X = \sum_{A \in \mathcal{F}} X_A$ is the total number of monochromatic members.

For a member $A$ to be monochromatic, all its $|A| = k$ points must receive the same color. Since the colors are assigned at random and independently, this implies that each member of $\mathcal{F}$ will be monochromatic with probability at most $2 \cdot 2^{-k} = 2^{1-k}$ (factor 2 comes since we have two colors). Hence,

$$\mathrm{E}[X] = \sum_{A \in \mathcal{F}} \mathrm{E}[X_A] = \sum_{A \in \mathcal{F}} 2^{1-k} = |\mathcal{F}| \cdot 2^{1-k}.$$

Since points in our probability space are 2-colorings, the pigeonhole property of expectation implies that a coloring, leaving at most $|\mathcal{F}| \cdot 2^{1-k}$ members of $\mathcal{F}$ monochromatic, must exist.

In particular, if $|\mathcal{F}| < 2^{k-1}$ then no member of $\mathcal{F}$ will be left monochromatic. □

The proof was quite easy. So one could ask whether we can replace $2^{k-1}$ by, say, $4^k$? By turning the probabilistic argument "on its head" it can be shown that this is *not* possible. The sets now become random and each coloring defines an event.

**Theorem 3.5** (Erdős 1964a)**.** *If k is sufficiently large, then there exists a k-uniform family* $\mathcal{F}$ *such that* $|\mathcal{F}| \leq k^2 2^k$ *and* $\mathcal{F}$ *is not* 2*-colorable.*

*Proof.* Set $r = \lfloor k^2/2 \rfloor$. Let $\boldsymbol{A}_1, \boldsymbol{A}_2, \ldots$ be independent random members of $\binom{[r]}{k}$, that is, $\boldsymbol{A}_i$ ranges over the set of all $A \subseteq \{1, \ldots, r\}$ with $|A| = k$, and $\Pr[\boldsymbol{A}_i = A] = \binom{r}{k}^{-1}$. Consider the family $\boldsymbol{\mathcal{F}} = \{\boldsymbol{A}_1, \ldots, \boldsymbol{A}_b\}$, where $b$ is a parameter to be specified later. Let $\chi$ be a coloring of $\{1, \ldots, r\}$ in red and blue, with $a$ red points and $r - a$ blue points. Using Jensen's inequality (see Proposition 1.12), for any such coloring and any $i$, we have

$$\begin{aligned}\Pr[\boldsymbol{A}_i \text{ is monochromatic}] &= \Pr[\boldsymbol{A}_i \text{ is red}] + \Pr[\boldsymbol{A}_i \text{ is blue}] \\ &= \frac{\binom{a}{k} + \binom{r-a}{k}}{\binom{r}{k}} \geq 2\binom{r/2}{k} \Big/ \binom{r}{k} := p,\end{aligned}$$

where, by the asymptotic formula (1.9) for the binomial coefficients, $p$ is about $\mathrm{e}^{-1}2^{1-k}$. Since the members $\boldsymbol{A}_i$ of $\boldsymbol{\mathcal{F}}$ are independent, the probability that a given coloring $\chi$ is legal for $\boldsymbol{\mathcal{F}}$ equals

$$\prod_{i=1}^{b} (1 - \Pr[\boldsymbol{A}_i \text{ is monochromatic}]) \leq (1-p)^b.$$

Hence, the probability that at least one of all $2^r$ possible colorings will be legal for $\boldsymbol{\mathcal{F}}$ does not exceed $2^r(1-p)^b < \mathrm{e}^{r\ln 2 - pb}$, which is less than 1 for $b = (r\ln 2)/p = (1+o(1))k^2 2^{k-2}\mathrm{e}\ln 2$. But this means that there must be at least one realization of the (random) family $\boldsymbol{\mathcal{F}}$, which has only $b$ sets and which cannot be colored legally. □

Let $B(k)$ be the minimum possible number of sets in a $k$-uniform family which is not 2-colorable. We have already shown that

$$2^{k-1} \leq B(k) \leq k^2 2^k\,.$$

As for exact values of $B(k)$, only the first two $B(2) = 3$ and $B(3) = 7$ are known. The value $B(2) = 3$ is realized by the graph $K_3$. We address the inequality $B(3) \leq 7$ in Exercise 3.9.

There is yet another class of 2-colorable families, without any uniformity restriction.

**Theorem 3.6.** *Let $\mathcal{F}$ be an arbitrary family of subsets of a finite set, each of which has at least two elements. If every two non-disjoint members of $\mathcal{F}$ share at least two common elements, then $\mathcal{F}$ is 2-colorable.*

*Proof.* Let $X = \{x_1, \ldots, x_n\}$ be the underlying set. We will color the points $x_1, \ldots, x_n$ one-by-one so that we do not color all points of any set in $\mathcal{F}$ with the same color. Color the first point $x_1$ arbitrarily. Suppose that $x_1, \ldots, x_i$ are already colored. If we cannot color the next element $x_{i+1}$ in red then this means that there is a set $A \in \mathcal{F}$ such that $A \subseteq \{x_1, \ldots, x_{i+1}\}$, $x_{i+1} \in A$ and all the points in $A \setminus \{x_{i+1}\}$ are red. Similarly, if we cannot color the next element $x_{i+1}$ in blue, then there is a set $B \in \mathcal{F}$ such that $B \subseteq \{x_1, \ldots, x_{i+1}\}$,

$x_{i+1} \in B$ and all the points in $B \setminus \{x_{i+1}\}$ are blue. But then $A \cap B = \{x_{i+1}\}$, a contradiction. Thus, we *can* color the point $x_{i+1}$ either red or blue. Proceeding in this way we will finally color all the points and no set of $\mathcal{F}$ becomes monochromatic. □

## 3.6 The choice number of graphs

The *choice number* (or *list-coloring number*) of a graph $G$, denoted by $ch(G)$, is the minimum integer $k$ such that for every assignment of a set $S(v)$ of $k$ colors to every vertex $v$ of $G$, there is a legal coloring of $G$ that assigns to each vertex $v$ a color from $S(v)$. Recall that a coloring is *legal* if adjacent vertices receive different colors. Obviously, this number is at least the chromatic number $\chi(G)$ of $G$.

**Theorem 3.7** (Alon 1992)**.** *For every bipartite $n \times n$ graph $G$ with $n \geq 3$, we have that $ch(G) \leq 2 \log_2 n$.*

*Proof.* Let $G = (V_0 \cup V_1, E)$ be a bipartite graph with $|V_0| = |V_1| = n$. Suppose that each vertex $v$ is assigned a set $S(v)$ of at least $2 \log_2 n$ colors, and let $S = \cup_v S(v)$ be the set of all colors. Since the graph is bipartite, it is enough to show that there is a partition $S = S_0 \cup S_1$ of $S$ such that

$$S_i \cap S(v) \neq \emptyset \text{ for both } i = 0, 1 \text{ and all } v \in V_i. \tag{3.4}$$

Then, for every two (even not necessarily adjacent) vertices $u \in V_0$ and $v \in V_1$, we can choose arbitrary colors $c_v \in S_0 \cap S(v)$ and $c_u \in S_1 \cap S(v)$; since $S_0 \cap S_1 = \emptyset$, these colors are clearly distinct.

To define such a partition $S = S_0 \cup S_1$ just flip, for each color $c \in S$, a fair 0-1 coin to decide whether to include this color in the set $S_0$; let also $S_1 = S \setminus S_0$. For a fixed $i \in \{0, 1\}$ and $v \in V_i$ we have that

$$\Pr[S_i \cap S(v) = \emptyset] = 2^{-|S(v)|} \leq 2^{-2 \log_2 n} = \frac{1}{n^2} < \frac{1}{2n}.$$

The number of pairs $(i, v)$ with $i \in \{0, 1\}$ and $v \in V_i$ is $2n$. Hence, by the union bound, the probability that our random partition $S = S_0 \cup S_1$ does not satisfy (3.4) is strictly smaller than 1, implying that a desired partition exists. □

Note that the proof says a bit more. If $A_1, \ldots, A_n$ and $B_1, \ldots, B_n$ are any two sequences of not necessarily distinct $2 \log_2 n$-element subsets of some set of vertices, then it is possible to color the vertices in red and blue so that each of the $A_i$ receives at least one red color and each of the $B_i$ receives at least one blue color.

## Exercises

**3.1.** Let $\Omega = B_1 \cup B_2$ be a partition of a sample space, and $A \subseteq \Omega$ be an event. Prove that then $\Pr[A]$ does not exceed the maximum of $\Pr[A \mid B_1]$ and $\Pr[A \mid B_2]$. *Hint*: Show that $\Pr[A] = \Pr[B_1] \cdot \Pr[A \mid B_1] + \Pr[B_2] \cdot \Pr[A \mid B_2]$.

**3.2.** Prove the following *Bonferroni inequality*:

$$\Pr[A_1 \cap \cdots \cap A_n] \geq \Pr[A_1] + \cdots + \Pr[A_n] - n + 1 \,.$$

**3.3.** Let $X, Y : \Omega \to \mathbb{R}$ be random variables. The *variance* of a random variable $X$ is defined as $\mathrm{Var}[X] := \mathrm{E}\left[(X - \mathrm{E}[X])^2\right]$. Prove that

1. $\mathrm{E}[a \cdot X + b \cdot Y] = a \cdot \mathrm{E}[X] + b \cdot \mathrm{E}[Y]$ for any constants $a$ and $b$.
2. If $X$ and $Y$ are independent then $\mathrm{E}[X \cdot Y] = \mathrm{E}[X] \cdot \mathrm{E}[Y]$ and $\mathrm{Var}[X + Y] = \mathrm{Var}[X] + \mathrm{Var}[Y]$.
3. $\mathrm{Var}[X] = \mathrm{E}[X^2] - \mathrm{E}[X]^2$. *Hint*: $\mathrm{E}[X \cdot \mathrm{E}[X]] = \mathrm{E}[X]^2$.

**3.4.** Let $X$ be a non-negative integer-valued random variable. Show that $\mathrm{E}[X^2] \geq \mathrm{E}[X]$, $\Pr[X = 0] \geq 1 - \mathrm{E}[X]$ and $\mathrm{E}[X] = \sum_{x=1}^{\infty} \Pr[X \geq x]$.

**3.5.** Use the Cauchy–Schwarz inequality $(\sum_{i=1}^{n} a_i b_i)^2 \leq (\sum_{i=1}^{n} a_i^2)(\sum_{i=1}^{n} b_i^2)$ to show that, for any random variable $X$, $\mathrm{E}[X]^2 \leq \mathrm{E}[X^2]$.

**3.6.** Let $X_1, \ldots, X_n$ be $n$ independent 0-1 random variables such that $\Pr[X_i = 1] = p_i$ and $\Pr[X_i = 0] = 1 - p_i$. Let $X = \sum_{i=1}^{n} X_i \bmod 2$. Prove that $\Pr[X = 1] = \frac{1}{2}\left[1 - \prod_i (1 - 2p_i)\right]$. *Hint*: Consider the random variable $Y = Y_1 \cdots Y_n$, where $Y_i = 1 - 2X_i$, and observe that $\mathrm{E}[Y] = 1 - 2 \cdot \Pr[Y = -1]$.

**3.7.** For a graph $G$ let, as before, $\mathrm{bc}(G)$ denote the smallest weight of a biclique covering of $G$. Show that if an $n$-vertex graph $G$ has no independent set of size larger than $\alpha$ then $\mathrm{bc}(G) \geq n \log_2(n/\alpha)$. *Hint*: Argue as in the proof of the lower bound in Theorem 3.3, and show that $\mathrm{E}[X] \leq \alpha$.

**3.8.** For a graph $G = (V, E)$, let $\mu_G$ be the minimum of $(a + b)/ab$ over all pairs of integers $a, b \geq 1$ such that $G$ contains a copy of a complete bipartite $a \times b$ graph $K_{a,b}$. Show that $\mathrm{bc}(G) \geq \mu_G \cdot |E|$.

**3.9.** Prove that $B(3) \leq 7$. That is, exhibit a family of seven 3-element sets which is not 2-colorable. *Hint*: Consider the Fano configuration (Fig. 12.1).

**3.10.** (Razborov 1990). Consider the family of all pairs $(A, B)$ of disjoint $k$-element subsets of $\{1, \ldots, n\}$. A set $Y$ *separates* the pair $(A, B)$ if $A \subseteq Y$ and $B \cap Y = \emptyset$. Prove that there exist $\ell = 2k4^k \ln n$ sets such that every pair $(A, B)$ is separated by at least one of them. *Hint*: Pick subsets $\boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_\ell$ of $\{1, \ldots, n\}$ randomly and independently, each with probability $2^{-n}$. Show that the probability that none of them separates a given pair $(A, B)$ is at most $\left(1 - 2^{-2k}\right)^\ell$ and use the counting sieve.

**3.11.** Let $X$ be a set of $n = kr$ points and consider their colorings $c\colon X \to \{1,\dots,k\}$ by $k$ colors. Say that such a coloring $c$ is *balanced* if each color is used for the same number of points, i.e., if $|c^{-1}(i)| = r$ for every color $i = 1,\dots,k$. Given a $k$-element set of points, say that it is *differently colored* if no two of its points get the same color. Prove that there exist $\ell = O(k\mathrm{e}^k \log n)$ balanced colorings $c_1,\dots,c_\ell$ such that every $k$-element subset of $X$ is differently colored by at least one of them. *Hint*: Consider independent copies $\boldsymbol{c}_1,\dots,\boldsymbol{c}_\ell$ of a balanced coloring $\boldsymbol{c}$ selected at random from the set of all $n!/(r!)^k$ such colorings. Show that for every $k$-element subset $S$ of $X$, $\boldsymbol{c}$ colors $S$ differently with probability $p = r^k \cdot \binom{n}{k}^{-1}$. Use the counting sieve to show that, with probability at least $1 - \binom{n}{k}(1-p)^\ell$, every $k$-element subset $S$ will be colored differently by at least one of $\boldsymbol{c}_1,\dots,\boldsymbol{c}_\ell$. Recall that $r = n/k$ and use Proposition 1.4 to show that this probability is nonzero for some $\ell = O(k\mathrm{e}^k \log n)$

**3.12.** (Khasin 1969). Consider the *$k$-threshold function* $T_k^n(x_1,\dots,x_n)$ which outputs 1 if and only if $x_1 + \cdots + x_n \geq k$. In Sect. 6.3.2 we will show that any depth-3 *Or-And-Or formula* for $T_k^n$ must have size exponential in $k$. What about the upper bounds? Use the previous exercise to show that $T_k^n$ can be computed by a monotone *Or-And-Or formula* of size $O(k\mathrm{e}^k n \log n)$. *Hint*: Each balanced $k$-coloring $c$ of $\{1,\dots,n\}$ gives us an And-Or formula $F_c = \bigwedge_{i=1}^k \bigvee_{c(j)=i} x_j$. Use the previous exercise to combine them into an Or-And-Or formula for $T_k^n$.

**3.13.** Let $\mathcal{F}$ be a family, each member of which has $\geq 3$ points and any two members share exactly one point in common. Suppose also that $\mathcal{F}$ is not 2-colorable. Prove that: (i) every point $x$ belongs to at least two members of $\mathcal{F}$, and (ii) any two points $x, y$ belong to at least one member of $\mathcal{F}$. *Hint*: (i) Take $x \in A \in \mathcal{F}$, color $A \setminus \{x\}$ red and the rest blue. (ii) Select sets $A, B$ such that $x \in A \setminus B$ and $y \in B \setminus A$; color $(A \cup B) \setminus \{x, y\}$ red and everything else blue.

**3.14.** (Lovász 1973). Let $\mathcal{F}$ be 3-uniform family on $n \geq 5$ points, in which each pair of points occurs in the same number of sets. Prove that $\mathcal{F}$ is not 2-colorable. *Hint*: Suppose there is a 2-coloring, count the members of $\mathcal{F}$ in two ways: by the monochromatic pairs contained in them and also by the bichromatic pairs contained in them. Let $n_1$ and $n_2$ denote the number of red and blue points, respectively, and let $a$ be the number of members of $\mathcal{F}$ containing a given pair of points. We have $a\binom{n_1}{2}$ sets in $\mathcal{F}$ containing a pair $\{x, y\}$ of red points, and $a\binom{n_2}{2}$ sets containing a blue pair of points. Hence, $|\mathcal{F}|$ is the sum of these two numbers. On the other hand, each set of $\mathcal{F}$ contains exactly two pairs $\{x, y\}$ where $x$ is blue and $y$ is red; so $2|\mathcal{F}| = an_1n_2$. Compare these numbers, and use the arithmetic-geometric mean inequality (1.16) to show that the equality can hold only if $n \leq 4$.